

First Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

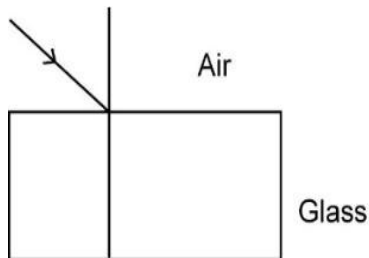
- A body of mass 2 kg is acted upon by two forces each of magnitude 2 N making an angle 120° with each other. Net acceleration of body is
a) Zero b) 1 m/s^2 c) $\sqrt{3} \text{ m/s}^2$ d) $2\sqrt{3} \text{ m/s}^2$
- What is the angle between $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$?
a) 0 b) $\pi/4$ c) $\pi/2$ d) π
- Identify the odd one:
a) displacement b) force c) speed d) acceleration
- The number of significant figures in the measurement $2.00 \times 10^{-2} \text{ m}$ is
a) 1 b) 2 c) 3 d) 4
- Which of the following is dimensional formula for specific heat capacity?
a) $[M^0 L^2 T^{-2} K^{-1}]$ b) $[M^0 L^2 T^{-3} K^{-1}]$
c) $[M^0 L T^{-2} K^{-1}]$ d) $[M L^2 T^{-2} K^{-1}]$
- If the value of coefficient of linear expansion of solid is 'n' times to that of its cubical expansion then the value of 'n' is
a) $1/2$ b) $1/3$ c) $2/3$ d) $3/2$
- When a convex mirror is completely immersed in the water, the focal length is
a) Increases b) decreases
c) remains same d) cannot be find then
- A concave mirror has the radius of curvature 20cm. an object is placed 10cm from the pole of the mirror. The image will be formed at
a) 10 cm from the pole on same side b) At infinity
c) 20cm from the pole on same side d) Will not be formed

9. What happens when a monochromatic light travels from air to glass
 - a) frequency and wavelength both decreases.
 - b) frequency increases and wavelength decreases.
 - c) frequency remains unchanged but wavelength decreases.
 - d) frequency and wavelength both remain unchanged.
10. If the velocity light in air is 3×10^8 m/s. The velocity of light in glass of refractive index 1.5 is:
 - a) 1.5×10^8 m/s
 - b) 2×10^8 m/s
 - c) 3×10^8 m/s
 - d) 4.5×10^8 m/s
11. A child is standing over the chair in an attempt to reach a coin at the bottom of water ($\mu = 1.33$) in the kitchen sink. If the depth of the water is 0.24m, what is the apparent depth of that coin:
 - a) 0.24 m
 - b) 0.31cm
 - c) 0.18m
 - d) 0.12m

Give short answer to the following questions.

12. a) Convert 1 dyne into Newton using dimensional analysis. [2]
b) Write the dimensional formula for coefficient of linear expansion. [1]
c) A student writes the expression for the escape velocity as $v = \sqrt{\frac{2GM}{R}}$, where symbols carry their usual meanings. Check the dimensional correctness of the formula. [2]
13. a) The force F is given in terms of time ' t ' and displacement ' x ' by the equation $F = A \sin Bx + C \sin Dt$. What is the dimension of D/B ? [2]
b) The Vander wall's equation for gas is $(P + \frac{a}{V^2})(V - b) = RT$, where P is pressure, V is volume and T is temperature. Determine the dimensions of 'a' and 'b'. [2]
c) Define significant figures in measurement with an example. [1]
14. a) A disoriented physics professor drives 3.25 km north, then 4.75 km west and then 1.50 km south. Find the magnitude and direction of the resultant displacement using the method of resolution. [3]
b) Can the resultant of two vectors equal in magnitude be equal to either vector in magnitude? Explain. [2]

15. a) Define absolute zero temperature [1]
 b) State and explain zeroth law of thermodynamics [2]
 c) The temperature of the melting point of ice and that of boiling point of water at normal atmospheric pressure are marked 40 and 200 respectively on a certain thermometer. What is the temperature in $^{\circ}\text{C}$ when the reading on this thermometer is 80. [2]
16. An iron plate of initial area A_1 undergoes superficial expansion when it is heated from temperature θ_1 to θ_2 .
 a) Show that $A_2 = A_1(1 + \beta\Delta\theta)$ where β is superficial expansivity. [2]
 b) Does the value of β depends on initial area of solid? Explain. [1]
 c) The volume of a metal sphere is increased by 1% of its original volume, when it is heated from 20°C to 222°C . Calculate the coefficient of linear, and cubical expansion of the metal. [2]
17. a) Complete the figure while the glass slab produces the lateral shift. [1]
 b) Derive the formula for the lateral shift. [2]
 c) Write the maximum and minimum value of lateral shift produced and show it in graph. [1]
 d) Find the lateral shift produced by a glass slab of thickness 5m and refractive index 1.5 when a ray incident on the glass slab with angle of incident 30° . [1]
18. a) Derive the relation between real depth and apparent depth when an object in water is viewed from air. [3]
 b) A man with height 100cm is laid along the axis of a convex mirror of focal length 25 cm, his legs being at distance of 50 cm from the pole of mirror. Calculate the size of image of man formed. [2]
19. a) Define the total internal reflection. [1]
 b) Write any two examples of total internal reflection. [1]
 c) Establish the relationship between critical angle and refractive index of the medium. [2]
 d) Calculate the critical angle for the water-glass interface. [1]
 ($\mu_w = 1.33$ and $\mu_g = 1.5$)



Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a) State the parallelogram law of vector addition. [1]
b) Using the parallelogram law of vector addition, find the magnitude and direction of the resultant of two vectors $\vec{P}$ and $\vec{Q}$ with an angle ' θ ' between them. [3]
c) Two dogs pull horizontally on the ropes attached to a post, the angle between the ropes is 60° . If dog A exerts a force of 270N on one rope and dog B exerts a force of 300N on other rope, find the magnitude of the resultant force and the angle made by resultant with dog A's rope. [3]
d) Under what condition does the resultant of two vectors be zero? Explain. [1]
21. a) A circular metal plate has a small hole at its center is cooled. What happens to the size of hole? Explain. [1]
b) For isotropic solid, show that its linear expansivity is approximately equal to one-third of superficial expansivity. [2]
c) Two bodies made of the same material have the same external dimensions and appearance, but one is solid and the other is hollow. If they are heated by equal amount of heat, is the overall volume expansion same? Explain. [2]
d) A brass pendulum clock keeps correct time at 15°C . How many seconds per day it will lose or gain at 0°C ?
(linear expansivity of brass = $20 \times 10^{-6} \text{ K}^{-1}$). [3]
22. a) Which mirror is shown in the figure? [1]
b) Define this mirror. [1]
c) Write any two uses of this mirror in daily life. [1]
d) Trace the property of the image formed by this mirror when a real object is placed nearer than its focal length. [2]
e) Derive the mirror formula for this mirror. [3]



❧ - The End - ❧